

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1259

Gardner 230 kV
July 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1259. The Interconnection Customer proposes a 262.4 MW wind facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on July 28, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 482 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,434,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Top Crop 500/230 kV XFMR	Grand Ridge Battery Projects to Gardner 230 KV	109.9	1,350	95.0
Gardner Substation to Grand Ridge Battery Projects 230 KV	Gardner to Top Crop 230 KV	108.0	1,521	99.9
Top Crop to Gardner 230 KV L	Gardner to Grand Ridge Battery Projects 230 KV	108.0	1,521	99.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Top Crop 500/230 kV XFMR	Replace transformer with higher-rated unit	\$20,512,000	16% (\$3,348,000)
Grand Ridge Battery Projects to Gardner 230 KV	Reconductor line	\$2,865,000	16% (\$468,000)
Top Crop to Gardner 230 KV	Reconductor line	\$3,789,000	16% (\$618,000)
	Total allocated to project		\$4,434,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.